Progress meeting

Optimisation-driven quantum circuit reduction

Reproduction, exact Clifford ground truth, and a general hardware-aware optimiser

Arnav Kapoor

IISER Bhopal

Study of B. Rosenhahn, T. J. Osborne, C. Hirche, *Optimization driven quantum circuit reduction*, New J. Phys. **27**, 104509 (2025), arXiv:2502.14715

The story in one slide

What I was asked to do

- Understand the paper, re-run every figure and table.
- Find its weaknesses.
- Extend it.

What I found

- It reproduces, but with qualifications.
- Every gate it uses is **Clifford**, so exact optima are computable.
- Against them, the method is **3–4.5**$\times$ from optimal.

The story in one slide

What I was asked to do

- Understand the paper, re-run every figure and table.
- Find its weaknesses.
- Extend it.

What I found

- It reproduces, but with qualifications.
- Every gate it uses is **Clifford**, so exact optima are computable.
- Against them, the method is **3–4.5**$\times$ from optimal.

What I built

- Theory of local optimality, parity and random benchmarks.
- A **general optimiser** for real native gates, connectivity and calibrated noise, with exact, state-aware or approximate semantics.

Where it stands

- Proven optimum on 35/40 of the paper's 3-qubit circuits.
- On Qiskit L3 (58 algorithms) it never adds 2q gates and removes 13 % of gates on Heron.
- BQSKit needs fewer 2q gates but is approximate and slower.
- Up to 95 qubits, it removes 19 % of gates on top of Qiskit L3 (Heron), each output with a replay certificate.
- Nam gate set (26 circuits), second to VOQC alone, and VOQC + ours best of all ($-25\,\%$).
- Paper drafted, every output verified.

Roadmap

1. **The paper.** Database-driven local rewriting
2. **Reproduction.** What holds and what needs qualifying
3. **Ground truth.** Every gate is Clifford $\Rightarrow$ exact optima
4. **Theory.** Exactness, parity, mixing, lower bounds
5. **A general optimiser.** No Clifford crutch, real hardware
6. **Results.** Paper benchmarks, standard suites, BQSKit, noisy simulation
7. **Wrap-up.** Lessons, caveats, critique, next steps

Part I

The paper

Rosenhahn, Osborne & Hirche shorten circuits by replacing short windows with shorter equivalents from a precomputed database.

Why shorten circuits at all?

- On NISQ hardware every gate costs fidelity. Roughly,

$$\Pr[\text{success}] \approx \prod_{\text{gates } g} \big(1 - e(g)\big),$$

  with two-qubit errors $\sim 10^{-3}$–10^{-2}, 10–100$\times$ single-qubit ones.

- Compiling to a device's **native alphabet** leaves redundancy. One textbook gate becomes several native ones, and neighbours stop cancelling.

- The goal is an equivalent circuit with fewer (or cheaper) native gates.

Figure from arXiv:2502.14715, Fig. 1

Why shorten circuits at all?

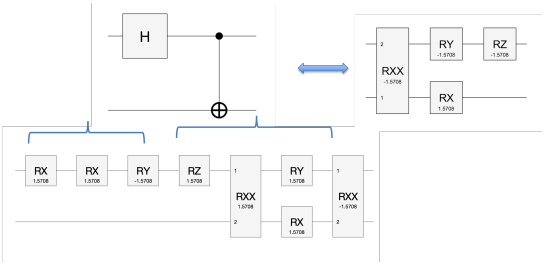
- On NISQ hardware every gate costs fidelity. Roughly,

$$\Pr[\text{success}] \approx \prod_{\text{gates } g} \big(1 - e(g)\big),$$

  with two-qubit errors $\sim 10^{-3}$–10^{-2}, 10–100$\times$ single-qubit ones.

- Compiling to a device's **native alphabet** leaves redundancy. One textbook gate becomes several native ones, and neighbours stop cancelling.

- The goal is an equivalent circuit with fewer (or cheaper) native gates.

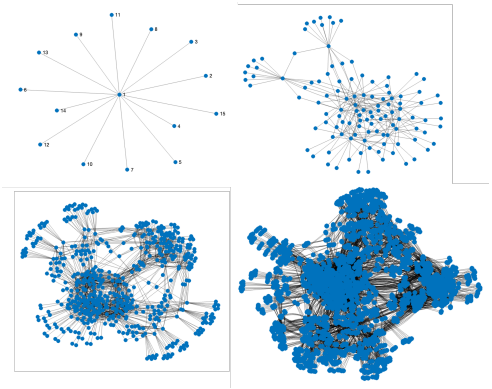


The paper's motivating example, H + CNOT compiled to the ion-trap alphabet (8 gates) and the shortest equivalent native word (4 gates). We confirmed by exhaustive search that no 3-gate version exists.

Figure from arXiv:2502.14715, Fig. 1

Step 1. The compute graph, a database of shortest circuits

- Fix a **discrete alphabet** on 2–3 qubits
  - ion trap, $R_x, R_y, R_z(\pm\frac{\pi}{2})$, $R_{XX}(\pm\frac{\pi}{2})$
  - IBM-style, $R_x, R_z(\pm\frac{\pi}{2})$, CZ
- Breadth-first search over words. Each node is a unitary (up to global phase), stored with its *shortest* word.
- Depth d = maximal word length enumerated.



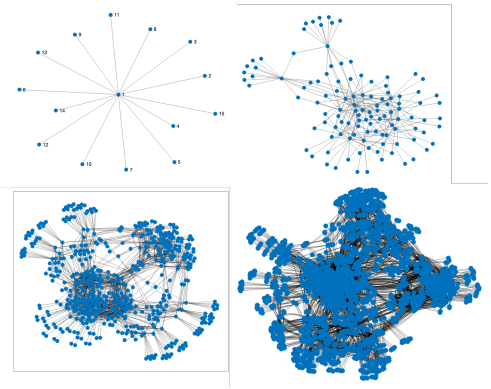
Compute graphs of growing depth (paper, Fig. 2).

Step 1. The compute graph, a database of shortest circuits

- Fix a **discrete alphabet** on 2–3 qubits
  - ion trap, $R_x, R_y, R_z(\pm\frac{\pi}{2})$, $R_{XX}(\pm\frac{\pi}{2})$
  - IBM-style, $R_x, R_z(\pm\frac{\pi}{2})$, CZ
- Breadth-first search over words. Each node is a unitary (up to global phase), stored with its *shortest* word.
- Depth d = maximal word length enumerated.

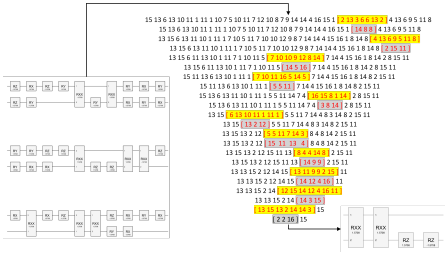
depth	1	2	3	4	5
2q, 14 ops	15	114	584	2,024	4,512
3q, 24 ops	25	337	3,215	23,622	137,572

Table I node counts. Our rebuild matches the paper exactly.



Compute graphs of growing depth (paper, Fig. 2).

Step 2. Slide a window, replace, repeat

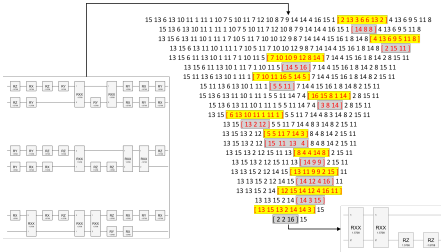


Circuit as a word of tokens, with windows replaced by shorter database words (paper, Fig. 3).

- Pick a window of 2–8 gates on ≤ 3 wires.
- Look its unitary up and substitute a shorter word if one exists.
- Shuffle commuting gates to expose new windows, and repeat until no progress.

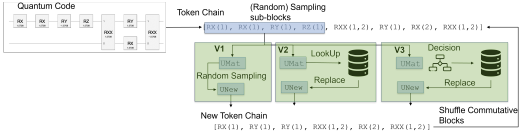
Figures from arXiv:2502.14715, Figs. 3 and 6

Step 2. Slide a window, replace, repeat



Circuit as a word of tokens, with windows replaced by shorter database words (paper, Fig. 3).

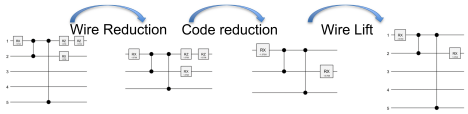
- Pick a window of 2–8 gates on ≤ 3 wires.
- Look its unitary up and substitute a shorter word if one exists.
- Shuffle commuting gates to expose new windows, and repeat until no progress.



- **V1.** Random sampling for a shorter equivalent word.
- **V2.** Database lookup.
- **V3.** A random forest first predicts whether a window is reducible, and only then is it looked up.

Figures from arXiv:2502.14715, Figs. 3 and 6

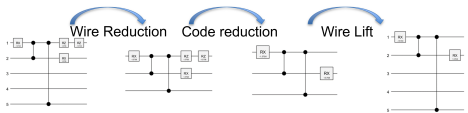
Step 3. Scaling up, and what the paper claims



Wire reduction, code reduction and wire lift apply a 3-qubit database inside an n-qubit circuit.

Takeaway. A clean, gate-set-agnostic idea, evaluated against Qiskit only, on random words over a discrete alphabet.

Step 3. Scaling up, and what the paper claims



Wire reduction, code reduction and wire lift apply a 3-qubit database inside an n-qubit circuit.

Claims

benchmark (random native circuits)	paper	Qiskit (theirs)
T. III, ion, 3q, 300 gates	45	176
T. IV, IBM-style, 3q, 300 gates	72	153
T. V, IBM-style, 15q, 500 gates	293	313
journal, 4q, 300 gates (ion / IBM)	111 / 107	–
T. II time (s), V1 / V2 / V3	199 / 55 / 38	

- Fig. 9 reports better results for reduced circuits on IBM Brisbane and Kyiv.

Takeaway. A clean, gate-set-agnostic idea, evaluated against Qiskit only, on random words over a discrete alphabet.

Part II

Reproduction

Re-running every figure and table with a Python port of the authors' own MATLAB code.

What I started from, and what I built

Inputs

- arXiv PDF + LaTeX source, journal (NJP) PDF.
- Official code `QCOptimDemo.zip`, linked only in the journal's ref. [38] (**V2 only**, MATLAB).
- Not specified in the paper are the database for Tables III–V, the Table II protocol, forest details, the transpiler level and the Qiskit version.

What I started from, and what I built

Inputs

- arXiv PDF + LaTeX source, journal (NJP) PDF.
- Official code `QCOptimDemo.zip`, linked only in the journal's ref. [38] (**V2 only**, MATLAB).
- Not specified in the paper are the database for Tables III–V, the Table II protocol, forest details, the transpiler level and the Qiskit version.

Built

- `qcr/`, a line-by-line Python port (graph builder, V1/V2/V3, wire reduction) with tests.
- An independent exact engine (Clifford tableaux) as a cross-check.
- Every output checked equivalent to its input (unitary error $\sim 10^{-12}$).
- Scripts and raw results for each table and figure, and a reproduction report (PDF).

Verdict at a glance

Claim	Status	Detail
Table I (compute graphs)	**exact**	identical node and edge counts
Figs. 1, 4, 5 (examples)	**yes**	same end points, but Fig. 4 needs a different gate pool (parity, Part IV)
Tables III/IV, journal 4q	**qualified**	only with the 24-operator, depth-5 database (unstated)
Table II, V1 $>$ V2 $>$ V3	**qualified**	V3 wins only with a tuned forest threshold
Table V (15 qubits)	**no longer**	ties a current Qiskit (2.5)
Fig. 9 (hardware)	**qualified**	real at Qiskit levels 0/1, disappears at level 3

Verdict at a glance

Claim	Status	Detail
Table I (compute graphs)	**exact**	identical node and edge counts
Figs. 1, 4, 5 (examples)	**yes**	same end points, but Fig. 4 needs a different gate pool (parity, Part IV)
Tables III/IV, journal 4q	**qualified**	only with the 24-operator, depth-5 database (unstated)
Table II, V1 $>$ V2 $>$ V3	**qualified**	V3 wins only with a tuned forest threshold
Table V (15 qubits)	**no longer**	ties a current Qiskit (2.5)
Fig. 9 (hardware)	**qualified**	real at Qiskit levels 0/1, disappears at level 3

Takeaway. The headline claim (beats Qiskit on few-qubit random circuits) holds, but the details around it need qualifying.

Table II, V1 vs. V2 vs. V3

variant	mean (s)	median (s)	reached	lookups
V1 random sampling	254.0	134.2	73	51,350
V2 database	46.4	6.5	96	7,904
V3 forest, $t = 0.5$	171.8	16.9	79	915
V3 forest, $t = 0.2$	33.7	5.3	98	1,216

100 random 5-qubit ion circuits, length 100 $\rightarrow$ target 60, cap 600 s. The paper reports 199 / 55 / 38 s.

Fig. 7 reproduction: 100 random 5-qubit circuits, length 100 → 60

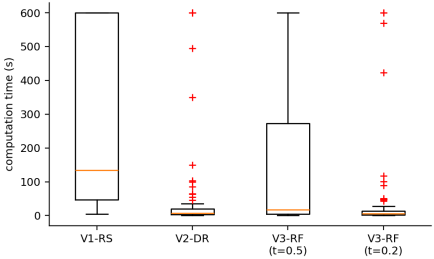
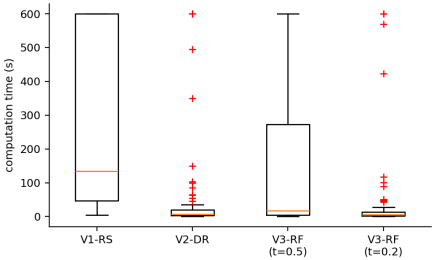


Table II, V1 vs. V2 vs. V3

variant	mean (s)	median (s)	reached	lookups
V1 random sampling	254.0	134.2	73	51,350
V2 database	46.4	6.5	96	7,904
V3 forest, $t = 0.5$	171.8	16.9	79	915
V3 forest, $t = 0.2$	33.7	5.3	98	1,216

100 random 5-qubit ion circuits, length 100 → target 60, cap 600 s. The paper reports 199 / 55 / 38 s.

Fig. 7 reproduction: 100 random 5-qubit circuits, length 100 → 60



- V1 ≫ V2 holds, so the database is a real speed-up.
- V3 depends entirely on the forest's **recall**. On held-out windows it flags only **30 %** of reducible ones at the default threshold 0.5 (92 % at 0.2).
- At 0.5 it discards most reductions and is *slower* than V2. The paper reports neither features, forest size, threshold nor accuracy.

Takeaway. The "learning" variant only helps after tuning a threshold the paper does not mention, and it saves time on a lookup that need not be slow.

Tables III–V and the journal statistics against Qiskit

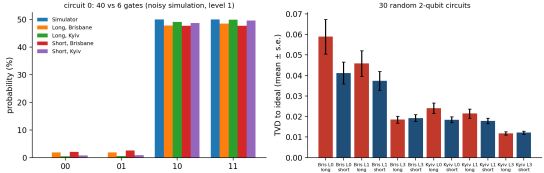
benchmark	input	Qiskit (paper)	Qiskit 2.5 L3	paper	our faithful port
T. III ion, 3q (20 circuits)	300	176	127.2	45	**47.1**
T. IV IBM-style, 3q (20)	300	153	104.0	72	**36.8**
journal ion, 4q (30)	300	–	164.9	111	**78.8**
journal IBM-style, 4q (30)	300	–	140.9	107	**76.0**
T. V IBM-style, 15q (10)	500	313	332.1	293	**335.9**

Tables III–V and the journal statistics against Qiskit

benchmark	input	Qiskit (paper)	Qiskit 2.5 L3	paper	our faithful port
T. III ion, 3q (20 circuits)	300	176	127.2	45	**47.1**
T. IV IBM-style, 3q (20)	300	153	104.0	72	**36.8**
journal ion, 4q (30)	300	–	164.9	111	**78.8**
journal IBM-style, 4q (30)	300	–	140.9	107	**76.0**
T. V IBM-style, 15q (10)	500	313	332.1	293	**335.9**

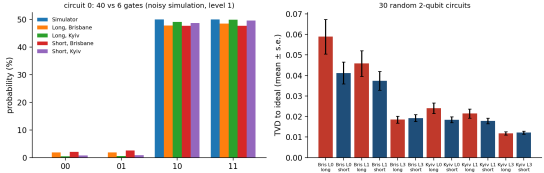
- With few qubits the method clearly beats Qiskit, as claimed, and our port is even better than published.
- But only with the larger database (24 operators, depth 5), which the paper does not state.
- Qiskit 2.5 is much stronger than the paper's Qiskit at 3 qubits (127 vs. 176 and 104 vs. 153).
- At 15 qubits our port reaches 335.9 (paper 293) and only ties Qiskit L3 (332.1, same CZ count). Random 15-qubit gate pairs rarely share wires, so 3-wire windows find little that Qiskit misses.

Fig. 9, the hardware experiment, in a noisy simulator



TVD to ideal (30 circuits)	level 0	level 1	level 3
Brisbane, long → reduced	0.059 → 0.041	0.046 → 0.037	0.018 vs. 0.019
Kyiv, long → reduced	0.024 → 0.018	0.021 → 0.018	0.012 vs. 0.012

Fig. 9, the hardware experiment, in a noisy simulator



TVD to ideal (30 circuits)	level 0	level 1	level 3
Brisbane, long → reduced	0.059 → 0.041	0.046 → 0.037	0.018 vs. 0.019
Kyiv, long → reduced	0.024 → 0.018	0.021 → 0.018	0.012 vs. 0.012

- At levels 0/1 the reduced circuit is closer to ideal (22/30 and 20/30 circuits), as in the paper.
- At level 3 Qiskit re-synthesises every 2-qubit block, both versions compile to ≈ 1.1 CZ, and the effect **vanishes**.
- Plain Qiskit L3 on the *long* circuit beats the reduced circuit at level 1 (0.018 vs. 0.037).
- The paper uses one circuit, one run per chip and no error bars, and does not state the transpiler level.

Bugs and unstated choices found along the way

In the released code

1. `simplereduce` never tests the last adjacent pair, and skips the pair after a deletion. The final circuit in the paper's Fig. 4 still contains an inverse pair (true optimum 2, shown 4).
2. The database excludes the identity, so identity blocks are never deleted by lookup.
3. There are three tolerances, 10^{-5} (paper), 10^{-3} (optimiser), 10^{-4} (graph builder).

Bugs and unstated choices found along the way

In the released code

1. `simplereduce` never tests the last adjacent pair, and skips the pair after a deletion. The final circuit in the paper's Fig. 4 still contains an inverse pair (true optimum 2, shown 4).
2. The database excludes the identity, so identity blocks are never deleted by lookup.
3. There are three tolerances, 10^{-5} (paper), 10^{-3} (optimiser), 10^{-4} (graph builder).

In the paper

4. Table I implies a pool of 14/24 operators, but the demo uses 13/21.
5. The database for Tables III–V is unstated, and only the larger one reproduces Table III.
6. Table II states neither the qubit count nor the target length, and the forest is unspecified.
7. For the hardware runs the text says RX, RY, CX but the tables say RX, RZ, CZ, and the transpiler level and shots are unstated.

Part III

Exact ground truth

The observation the paper misses. Every gate it uses is Clifford, so the search space is a finite group we can enumerate completely.

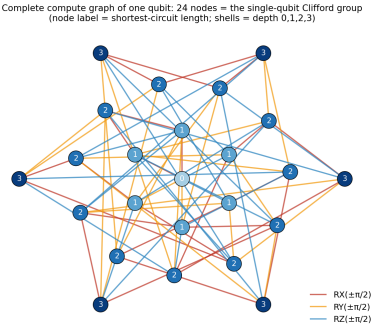
Key observation, every gate in the paper is Clifford

- $\pm\frac{\pi}{2}$ rotations, $R_{XX}(\pm\frac{\pi}{2})$ and CZ map Paulis to Paulis.
- So every circuit is an element of the finite **Clifford group** $\mathcal{C}_n$ (mod phase), of order

$$2^{n^2+2n}\prod_{j=1}^{n}(4^j-1)$$

(24 for $n=1$, 11,520 for $n=2$, **92,897,280** for $n=3$).

- The paper's "compute graph" is a ball in the **Cayley graph** of this group.



For one qubit the complete compute graph has exactly 24 nodes and diameter 3.

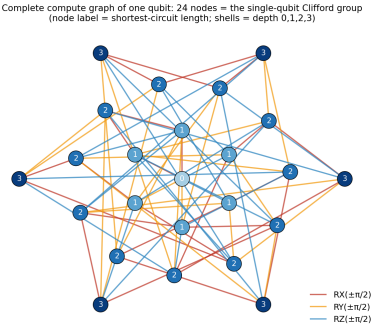
Key observation, every gate in the paper is Clifford

- $\pm\frac{\pi}{2}$ rotations, $R_{XX}(\pm\frac{\pi}{2})$ and CZ map Paulis to Paulis.
- So every circuit is an element of the finite **Clifford group** $\mathcal{C}_n$ (mod phase), of order

$$2^{n^2+2n}\prod_{j=1}^{n}(4^j-1)$$

(24 for $n=1$, 11,520 for $n=2$, **92,897,280** for $n=3$).

- The paper's "compute graph" is a ball in the **Cayley graph** of this group.



For one qubit the complete compute graph has exactly 24 nodes and diameter 3.

Takeaway. Optimality is no longer a vague question. Each circuit has an exact shortest length, computable by search over one finite graph.

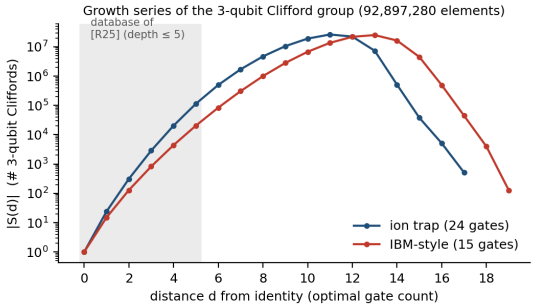
Enumerating all 92.9 million 3-qubit Cliffords

- A Clifford mod phase is its signed **tableau**, which for $n \leq 3$ packs into one 64-bit word.
- Gates act row-wise through lookup tables, so each BFS level is a vectorised gather. Exact membership, no tolerance.
- Full BFS takes 25 min and $\approx$6 GB per alphabet.

Enumerating all 92.9 million 3-qubit Cliffords

- A Clifford mod phase is its signed **tableau**, which for $n \leq 3$ packs into one 64-bit word.
- Gates act row-wise through lookup tables, so each BFS level is a vectorised gather. Exact membership, no tolerance.
- Full BFS takes 25 min and ≈6 GB per alphabet.

alphabet	n	diameter	mean optimal length
ion trap	2	10	5.89
IBM-style	2	11	6.92
ion trap	3	**17**	**10.72**
IBM-style	3	**19**	**12.34**



The paper's depth-5 database (shaded) covers 0.15 % of the group, and almost all elements sit at distance 10–14.

How far from optimal is the paper's method?

3 qubits, 300 random gates, 20 circuits	**proven optimum**	paper's method (our port)	Qiskit 2.5 L3	paper (1 circuit)
ion trap	**10.4**	47.1	127.2	45
IBM-style	**12.2**	36.8	104.0	72

How far from optimal is the paper's method?

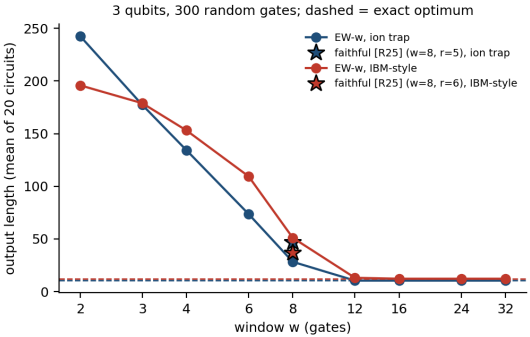
3 qubits, 300 random gates, 20 circuits	**proven optimum**	paper's method (our port)	Qiskit 2.5 L3	paper (1 circuit)
ion trap	**10.4**	47.1	127.2	45
IBM-style	**12.2**	36.8	104.0	72

The gap

The method's output is **4.5**× (ion trap) and **3.0**× (IBM-style) the optimum, on its own benchmark. Its showcase Fig. 4 output is 2× optimal.

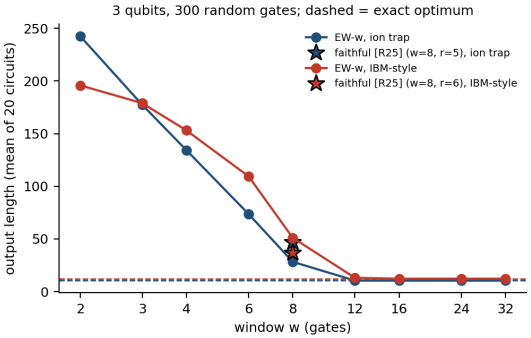
- Our port matches the paper's own single-circuit number (47.1 vs. 45), so this is the method, not the implementation.
- Beating Qiskit by 2.7× looked impressive, yet another 4.5× was available.

Exact windows close the gap



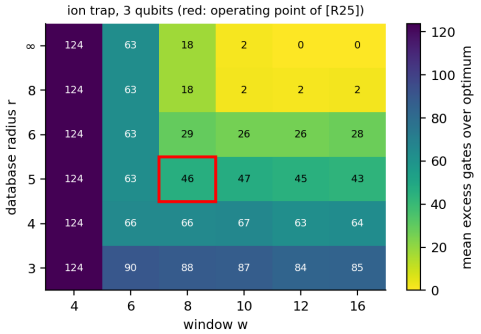
- **Exact-window optimiser** EW-w replaces every window of $\leq w$ gates on ≤ 3 wires by its *proven* optimal word (table lookup).

Exact windows close the gap



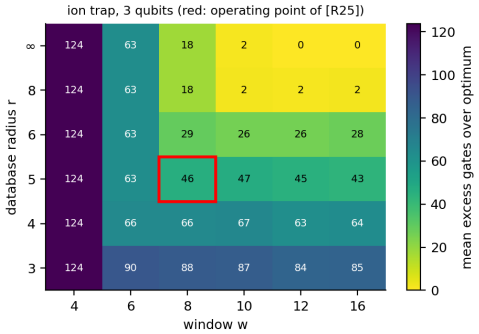
- **Exact-window optimiser** EW-w replaces every window of $\leq w$ gates on ≤ 3 wires by its *proven* optimal word (table lookup).
- The transition is sharp. $w \geq 12$ (ion) / $w \geq 16$ (IBM-style) reaches the **proven optimum on all 40 circuits**.
- 0.1–0.2 s per circuit, vs. 51 s / 17 s for the paper's method.
- Why so sharp? Once the circuit is shorter than the window, one more sweep is exact (the *cascade*, Part IV).

What limits the original method, window or database?



Mean excess over the optimum (ion trap, 3 qubits) by window w and database radius r. The paper's $(8, 5)$ is marked in red.

What limits the original method, window or database?



Mean excess over the optimum (ion trap, 3 qubits) by window w and database radius r. The paper's $(8, 5)$ is marked in red.

- A larger window at $r = 5$ lowers the excess from 46 to 43 and barely helps.
- A larger database at $w = 8$ lowers it from 46 to 18.
- With $w \geq 12$ and $r \geq 8$ the excess is ≤ 2.

Takeaway. The **database radius**, not the window or the search strategy, is the binding constraint. The theory predicts both are needed.

Four qubits, the journal benchmark

4q, 300 gates, 100 circuits	lower bound	CW-sem	CW-24	contiguous EW-16	port	published	Qiskit L3	BQSKit L4
ion trap, total	$\geq$11.1	**45.6**	56.7	110.3	78.8	111	164.9	120.6
IBM-style, total	$\geq$12.9	**49.3**	59.1	130.9	76.0	107	140.9	165.2
RXX (ion) / CZ (IBM)	–	–	27.6 / 27.1	44.2 / 47.1	33.3 / 32.4	43 / 43	47.3 / 47.3	27.6 / 35.1

Port, Qiskit L3 and BQSKit L4 use the first 30 circuits. BQSKit L4 is our run, approximate (error $< 10^{-7}$), all-to-all.

Four qubits, the journal benchmark

4q, 300 gates, 100 circuits	lower bound	CW-sem	CW-24	contiguous EW-16	port	published	Qiskit L3	BQSKit L4
ion trap, total	$\geq$11.1	**45.6**	56.7	110.3	78.8	111	164.9	120.6
IBM-style, total	$\geq$12.9	**49.3**	59.1	130.9	76.0	107	140.9	165.2
RXX (ion) / CZ (IBM)	–	–	27.6 / 27.1	44.2 / 47.1	33.3 / 32.4	43 / 43	47.3 / 47.3	27.6 / 35.1

- Same exact 3-qubit oracle, different windows. Contiguous windows (EW-16) are *no better* than published, **DAG-convex windows** (CW-24) cut the published results by 49 % / 45 %, and adaptive commutation-aware windows (CW-sem) go further.
- **Against BQSKit L4** on the same 30 circuits, CW-24 has about the same RXX count (27.0 vs. 27.6) and fewer CZ (28.1 vs. 35.1, fewer on 25/30), with 0.46× / 0.37× of its total gates, without targeting 2q gates.
- No exact optimum at 4 qubits ($|G| = 1.2 \times 10^{13}$), but a certified lower bound, and the remaining gap is at most $\approx 4\times$.

Port, Qiskit L3 and BQSKit L4 use the first 30 circuits. BQSKit L4 is our run, approximate (error $< 10^{-7}$), all-to-all.

Part IV

Theory

Local rewriting as rewriting on a Cayley graph. When is it exact, what can it never do, and what do random benchmarks measure?

The model

- Alphabet Γ, group $G = \langle \Gamma \rangle$, word length $|g|$, ball $B(r)$, diameter D.
- Compute graph of depth d = ball $B(d)$ in $\mathrm{Cay}(G, \Gamma)$, and stored words are geodesics.
- The rewriting system $\mathcal{R}(w, r)$ replaces a factor of length $\leq w$ by an equivalent geodesic of length $\leq r$.

The model

- Alphabet Γ, group $G = \langle \Gamma \rangle$, word length $|g|$, ball $B(r)$, diameter D.
- Compute graph of depth d = ball $B(d)$ in $\mathrm{Cay}(G, \Gamma)$, and stored words are geodesics.
- The rewriting system $\mathcal{R}(w, r)$ replaces a factor of length $\leq w$ by an equivalent geodesic of length $\leq r$.
- Swapping commuting neighbours = moves in the **trace monoid**.
- Factors of trace-equivalent words = **convex** sets of the circuit DAG.
- So the paper's random commuting shuffles are a *stochastic enumeration of convex windows*, and we enumerate them directly.
- The paper's V2 operates at $(w, r) = (8, 5)$.

Theorem 1. Local = global above the diameter

Exactness threshold

Let G be finite with diameter D. If $w \geq D + 1$ and $r \geq D$, every (w, r)-reduced word is a geodesic.

Proof idea. A reduced word longer than D would have a first factor of length $D + 1$, which is not geodesic and fits in the window. So the word has length $\leq D < w$, and then the whole word is one window.

Theorem 1. Local = global above the diameter

Exactness threshold

Let G be finite with diameter D. If $w \geq D + 1$ and $r \geq D$, every (w, r)-reduced word is a geodesic.

Proof idea. A reduced word longer than D would have a first factor of length $D + 1$, which is not geodesic and fits in the window. So the word has length $\leq D < w$, and then the whole word is one window.

Cascade

If at any time the current length $\ell \leq w$, the next sweep returns a geodesic. The relevant window is $w^\star = \min\{w : \ell(w) \leq w\}$.

- Measured $w^\star \approx 12$ (ion) / 16 (IBM-style) for 3 qubits, well below $D + 1 = 18$.
- The paper's $(8, 5)$ is below both thresholds.
- For continuous gate sets $D = \infty$ and no finite window is exact. Hence ground truth must come from the discrete case.

Parity

(a) For $n \leq 2$ the Cayley graph is **bipartite**, so every equivalent rewrite removes an *even* number of gates.
(b) For $n \geq 3$, with any generating set containing square roots of each X_i, Z_i, it is **never** bipartite.

Proof idea. Bipartite $\iff$ a sign character χ with $\chi(\Gamma) = -1$. For $n \geq 3$, χ kills the Paulis and factors through $\mathrm{Sp}(2n, 2)$, which is simple, so $\chi \equiv 1$.

Parity

(a) For $n \leq 2$ the Cayley graph is **bipartite**, so every equivalent rewrite removes an *even* number of gates.
(b) For $n \geq 3$, with any generating set containing square roots of each X_i, Z_i, it is **never** bipartite.

Proof idea. Bipartite $\iff$ a sign character χ with $\chi(\Gamma) = -1$. For $n \geq 3$, χ kills the Paulis and factors through $\mathrm{Sp}(2n, 2)$, which is simple, so $\chi \equiv 1$.

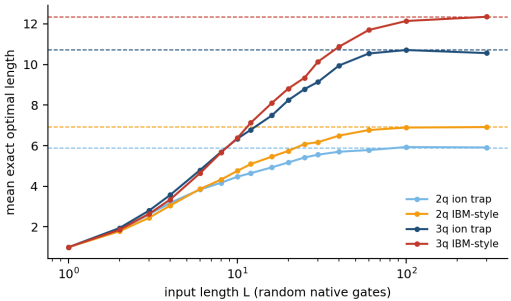
- Exhaustive search at 3 qubits finds a shortest odd relation of length $\rho = 11$ (ion) / 9 (IBM-style). Changing parity needs a window of ≥ 6 gates, and smaller windows can never reach an optimum of the other parity.
- The paper's Fig. 4 shows a $6 \to 3$ replacement on two qubits, which is **impossible** with the stated gate set.

Theorem 3. Random-circuit benchmarks measure a constant

Benchmarks mix

A random word of length L is within $\frac{1}{2}\sqrt{|G|}\,\beta^L$ (total variation) of a uniform group element. So $\mathbb{E}[\text{optimal length}] \to$ the mean distance in G.

For 2 qubits (ion trap) $\beta = 0.937$ (sparse diagonalisation).

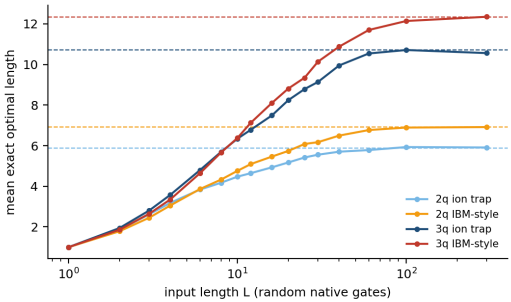


Theorem 3. Random-circuit benchmarks measure a constant

Benchmarks mix

A random word of length L is within $\frac{1}{2}\sqrt{|G|}\,\beta^L$ (total variation) of a uniform group element. So $\mathbb{E}[\text{optimal length}] \to$ the mean distance in G.

For 2 qubits (ion trap) $\beta = 0.937$ (sparse diagonalisation).



Takeaway. "Reduced from 300 to 45 gates" mostly measures the input length. Beyond $L \approx 100$, every random 3-qubit ion circuit has optimum ≈ 10.7 on average, whatever L is.

Lower bounds for every method (4 qubits)

Counting bound

$\Pr[|g| \leq d] \leq |B(d)|/|G|$, and spheres are submultiplicative, $|S(a+b)| \leq |S(a)||S(b)|$. Exact spheres up to a small depth bound $\mathbb{E}|g|$ from below for any optimiser.

Lower bounds for every method (4 qubits)

Counting bound

$\Pr[|g| \le d] \le |B(d)|/|G|$, and spheres are submultiplicative, $|S(a+b)| \le |S(a)||S(b)|$. Exact spheres up to a small depth bound $\mathbb{E}|g|$ from below for any optimiser.

- Check at $n = 3$, bound 7.9 vs. truth 10.72.
- $n = 4$ (1.2×10^{13} elements), spheres to depth 6 give $\mathbb{E}|g| \ge$ **11.1** (ion) / **12.9** (IBM-style).
- The best achieved is 45.6 / 49.3. Closing that gap is an open problem the framework makes precise.

Part V

A general optimiser

Drop the Clifford crutch and add continuous angles, connectivity, calibrated noise, and a precise notion of “equivalent”.

Why go beyond Clifford?

The Clifford results don't transfer

- Real compilers emit **arbitrary angles** ($R_z(\theta)$, MS(θ)), so there is no finite group and no table.
- Real chips have **connectivity** (heavy-hex) and per-gate **calibrated errors**.
- "Up to global phase" is **unsafe** once a block is controlled.
- Total gate count is the wrong objective when 2q gates dominate error.

Why go beyond Clifford?

The Clifford results don't transfer

- Real compilers emit **arbitrary angles** ($R_z(\theta)$, MS(θ)), so there is no finite group and no table.
- Real chips have **connectivity** (heavy-hex) and per-gate **calibrated errors**.
- "Up to global phase" is **unsafe** once a block is controlled.
- Total gate count is the wrong objective when 2q gates dominate error.

Design decisions

- **No** Clifford assumption in the method. Clifford tables are used only to *evaluate*.
- Devices are IBM Heron (FakeTorino with $R_z(\theta)$, $\sqrt{X}$, X, CZ, heavy-hex, calibrated) and a trapped-ion model ($R_{x,y,z}(\theta)$, $R_{XX}(\theta)$, all-to-all).
- Semantics are exact incl. phase, state/measurement-aware, or approximate with tracked error.
- The objective is chosen by noisy simulation, not assumed.

One idea, equivalence = group action and key = canonical orbit

notion	$U \sim V$ **iff**	**canonical database key**
exact	$V = U$	raw matrix
up to phase (the paper)	$V = e^{i\varphi}U$	phase-normalised matrix
input-aware (at start)	$V\|0\rangle = e^{i\varphi}U\|0\rangle$	phase-normalised $U\|0\rangle$
output-aware (before readout)	$V = DU$, D diagonal	rows phase-normalised
symmetry	$V = PUP^\dagger$, P a chip symmetry	minimum over the orbit
approximate	$\|U - V\| \leq \varepsilon$	exact key + verified distance

One idea, equivalence = group action and key = canonical orbit

notion	$U \sim V$ iff	canonical database key
exact	$V = U$	raw matrix
up to phase (the paper)	$V = e^{i\varphi}U$	phase-normalised matrix
input-aware (at start)	$V\|0\rangle = e^{i\varphi}U\|0\rangle$	phase-normalised $U\|0\rangle$
output-aware (before readout)	$V = DU$, D diagonal	rows phase-normalised
symmetry	$V = PUP^\dagger$, P a chip symmetry	minimum over the orbit
approximate	$\|U - V\| \leq \varepsilon$	exact key + verified distance

One engine, many semantics

The database stores 64-bit hashes of canonical forms, so one lookup finds the whole class. New semantics = new key function. Every hit is re-verified numerically.

Proposition 1 (state- and measurement-aware rewriting)

The program prepares $|0^n\rangle$, runs the circuit and measures all qubits. Let S be a convex window on wires W. If (a) nothing precedes S on W and $V|0_W\rangle = e^{i\varphi}U|0_W\rangle$, or (b) nothing follows S on W and $V = DU$ with D diagonal, then replacing U by V leaves the output distribution unchanged, and so does any sequence of such rewrites.

Why. In (a) the input is a product state, and in (b) $|\langle x|D\psi\rangle|^2 = |\langle x|\psi\rangle|^2$. The invariant is the output distribution.

Proposition 1 (state- and measurement-aware rewriting)

The program prepares $|0^n\rangle$, runs the circuit and measures all qubits. Let S be a convex window on wires W. If (a) nothing precedes S on W and $V|0_W\rangle = e^{i\varphi}U|0_W\rangle$, or (b) nothing follows S on W and $V = DU$ with D diagonal, then replacing U by V leaves the output distribution unchanged, and so does any sequence of such rewrites.

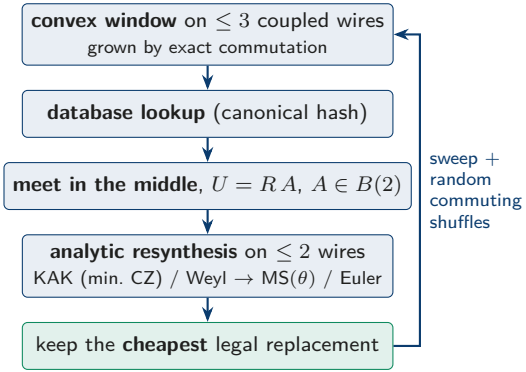
Why. In (a) the input is a product state, and in (b) $|\langle x|D\psi\rangle|^2 = |\langle x|\psi\rangle|^2$. The invariant is the output distribution.

Proposition 2 (approximate rewriting)

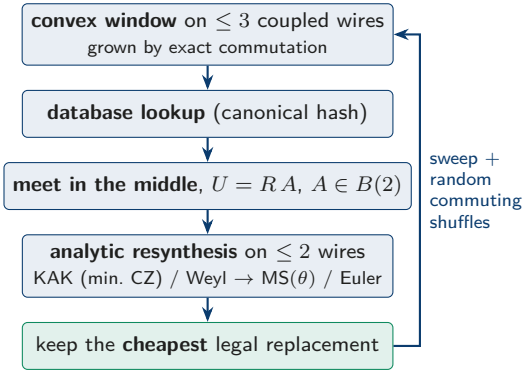
If rewrite i has operator-norm error ε_i, the final circuit is within $\sum_i \varepsilon_i$ of the original (including phase in exact mode).

Why. Telescoping, and $\|A(X-Y)B\| = \|X-Y\|$ for unitary A, B. The optimiser reports this bound with every circuit.

The engine


convex window on ≤ 3 coupled wires
grown by exact commutation
database lookup (canonical hash)
meet in the middle, $U = R\,A$, $A \in B(2)$
analytic resynthesis on ≤ 2 wires
KAK (min. CZ) / Weyl $\rightarrow$ MS(θ) / Euler
keep the **cheapest** legal replacement
sweep + random commuting shuffles

The engine



device, topology	notion	classes	depth
Heron, line	phase	6.31 M	9
	output-aware	0.34 M	10
	input-aware	1.41 M	14
ion, triangle	phase	28.4 M	6
	output-aware	7.36 M	6
	input-aware	31.6 M	7
paper ion alphabet	phase	2.93 M	9
paper IBM alphabet	phase	1.84 M	10

Grid angles $\pm\frac{\pi}{4}, \pm\frac{\pi}{2}, \pi$ give 23 (Heron) / 57 (ion) local gates. Builds take 18 s to 38 min. Deepest level stored as keys only.

Global phase and cost, the details that matter

Exact including global phase

- Inside windows, match up to phase and record each phase.
- Native gates reach only a *discrete* set of phases, multiples of $\pi/4$ (Heron) and $\pi/2$ (ion). Snap, or reject.
- Repair once at the end with the cheapest gadget. On ions absorb $\pm i$ into an existing MS gate, $R_{XX}(\theta \pm \pi)\, R_x(\pi)^{\otimes 2} = \pm i\, R_{XX}(\theta)$.
- Never return a costlier circuit than the input.

Global phase and cost, the details that matter

Exact including global phase

- Inside windows, match up to phase and record each phase.
- Native gates reach only a *discrete* set of phases, multiples of $\pi/4$ (Heron) and $\pi/2$ (ion). Snap, or reject.
- Repair once at the end with the cheapest gadget. On ions absorb $\pm i$ into an existing MS gate, $R_{XX}(\theta \pm \pi)\, R_x(\pi)^{\otimes 2} = \pm i\, R_{XX}(\theta)$.
- Never return a costlier circuit than the input.

Three candidate objectives

$$c_{\text{count}} = 1, \qquad c_{w2} = \begin{cases} 10 & \text{2q} \\ 1 & \text{1q} \end{cases}$$

$$c_{\text{cal}}(g) = -\log\bigl(1 - e(g)\bigr)$$

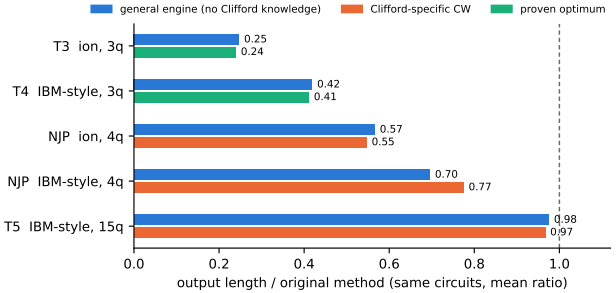
- $e^{-c_{\text{cal}}}$ = estimated success probability (ESP), so minimising c_{cal} maximises ESP.
- $e(g)$ from FakeTorino calibration per qubit / pair, the ion model uses $3 \cdot 10^{-5}$ / 10^{-3}, and R_z is virtual.
- Which one wins is decided by noisy simulation (Part VI).

Part VI

Results

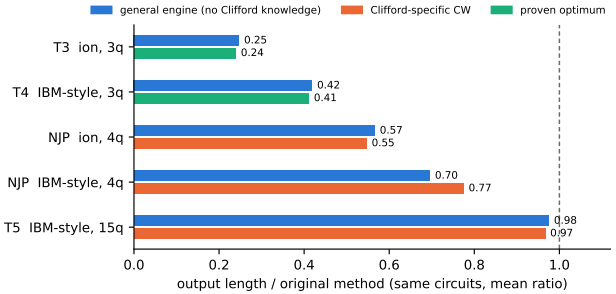
First the paper's own benchmarks against ground truth, then 60 standard circuits on hardware models, against Qiskit, TKET and BQSKit.

Result 1. The paper's benchmarks, without Clifford knowledge



Output length / paper's method, same circuits and seeds, paper semantics (up to phase, gate count, $\pm\frac{\pi}{2}$ alphabets).

Result 1. The paper's benchmarks, without Clifford knowledge



Output length / paper's method, same circuits and seeds, paper semantics (up to phase, gate count, $\pm\frac{\pi}{2}$ alphabets).

- **3 qubits.** Proven optimum on **17/20** (ion) and **18/20** (IBM-style), mean 10.8 vs. optimum 10.4.
- **4 qubits.** On par with the Clifford-specific optimiser (46.1 vs. 45.6 and 51.4 vs. 49.3).
- **15 qubits.** All methods within 4 %.
- 3–9 s per circuit, and all 250 outputs verified and native.

Takeaway. Knowing the group structure is not what makes the difference. Large databases and convex windows are.

Result 2. Standard suites on hardware models, setup

Circuits (60 per device, 2–12 qubits, $\leq$1,008 gates)

- Nam/VOQC arithmetic (Toffoli/Barenco ladders, adders, modular multiplication, $\mathrm{GF}(2^4)$ multiplier, product formulas, QFT8)
- QASMBench small/medium (4 with classical feed-forward and 4 unparsable skipped)
- QFT, QAOA, Heisenberg Trotter, EfficientSU2, Steane encoder
- Random non-Clifford circuits

Result 2. Standard suites on hardware models, setup

Circuits (60 per device, 2–12 qubits, $\leq$1,008 gates)

- Nam/VOQC arithmetic (Toffoli/Barenco ladders, adders, modular multiplication, $\mathrm{GF}(2^4)$ multiplier, product formulas, QFT8)
- QASMBench small/medium (4 with classical feed-forward and 4 unparsable skipped)
- QFT, QAOA, Heisenberg Trotter, EfficientSU2, Steane encoder
- Random non-Clifford circuits

Pipelines (same native set, coupling map, seed)

- Qiskit transpile, levels 1 and 3
- TKET `FullPeepholeOptimise` on L1 + rebase
- BQSKit 1.2.1, levels 2 and 3 on L1 (approximate resynthesis)
- ours on L1 output and **ours on L3 output**

Metrics. 2q count, total, depth, ESP and noisy-simulation TVD (≤ 8 active qubits). Every output is verified, native and coupled.

Result 2. Relative to Qiskit level 3

	IBM Heron (58 circuits)			**trapped ion** (58 circuits)	
pipeline	2q gates	all gates	ESP	2q gates	all gates
TKET	×1.02	×1.71	×1.16	×1.27	×1.79
ours, exact, from Qiskit L1	×1.00	**×0.84**	**×1.21**	**×0.95**	**×0.76**
ours, exact, from Qiskit L3	**×0.98**	**×0.87**	×1.03	×1.00	×1.00
ours, approximate ($\varepsilon = 10^{-3}$), from L1	×1.01	**×0.84**	**×1.21**	**×0.95**	**×0.75**
ours, state/measurement-aware	**×0.77**	**×0.61**	**×1.28**	**×0.62**	**×0.50**
BQSKit L2 on L1 (approximate)	×0.89	×1.26	**×1.21**	×0.96	**×0.74**
BQSKit L3 on L1 (approximate)	**×0.85**	×1.20	**×1.22**	**×0.94**	**×0.71**
ours, exact, on BQSKit L3	**×0.85**	×1.01	**×1.25**	**×0.94**	**×0.70**

Seed 0. Over 5 seeds ESP from L1 is $\times 1.10 \pm 0.11$ (Result 5). Ratios of summed counts, geometric mean for ESP. Nam pf1_10/pf2_10 are excluded because Qiskit L3 drops their 10^{-7} rad rotations (approximate, not exact).

Result 2. Relative to Qiskit level 3

	IBM Heron (58 circuits)			**trapped ion** (58 circuits)	
pipeline	2q gates	all gates	ESP	2q gates	all gates
TKET	×1.02	×1.71	×1.16	×1.27	×1.79
ours, exact, from Qiskit L1	×1.00	**×0.84**	**×1.21**	**×0.95**	**×0.76**
ours, exact, from Qiskit L3	**×0.98**	**×0.87**	×1.03	×1.00	×1.00
ours, approximate ($\varepsilon = 10^{-3}$), from L1	×1.01	**×0.84**	**×1.21**	**×0.95**	**×0.75**
ours, state/measurement-aware	**×0.77**	**×0.61**	**×1.28**	**×0.62**	**×0.50**
BQSKit L2 on L1 (approximate)	×0.89	×1.26	**×1.21**	×0.96	**×0.74**
BQSKit L3 on L1 (approximate)	**×0.85**	×1.20	**×1.22**	**×0.94**	**×0.71**
ours, exact, on BQSKit L3	**×0.85**	×1.01	**×1.25**	**×0.94**	**×0.70**

- **Stacked on L3** ours never adds 2q gates (Heron 9 better and 49 equal, ion 1 and 57) and never lowers ESP.
- TKET keeps 2q counts close to L3 but needs 1.7–1.8× the gates after rebasing.
- BQSKit L3 has the fewest 2q gates, with approximate outputs. Ours on top of it gives the best ESP (next slide).
- Ion ESPs differ by $< 1\,\%$ across pipelines (MS error 10^{-3}), so they are omitted.

Seed 0. Over 5 seeds ESP from L1 is ×1.10 ± 0.11 (Result 5). Ratios of summed counts, geometric mean for ESP. Nam pf1_10/pf2_10 are excluded because Qiskit L3 drops their 10^{-7} rad rotations (approximate, not exact).

Result 3. BQSKit, the strongest baseline

Summed over 58 circuits (2q / all gates)

	Heron	ion
Qiskit L3	2,889 / 15,404	1,378 / 7,425
ours from L1	2,902 / **12,908**	1,305 / 5,608
ours from L3	2,830 / 13,405	1,376 / 7,420
BQSKit L3	**2,446** / 18,510	1,302 / 5,250
ours on BQSKit L3*	2,448 / 15,630	**1,298** / **5,202**

Same Qiskit L1 input, native gates and couplers. BQSKit 1.2.1 at default settings, output mapped back to the physical qubits. *On a separate BQSKit L3 run (2,459 / 18,601 and 1,301 / 5,276 before our pass).

Result 3. BQSKit, the strongest baseline

Summed over 58 circuits (2q / all gates)

	Heron	ion
Qiskit L3	2,889 / 15,404	1,378 / 7,425
ours from L1	2,902 / **12,908**	1,305 / 5,608
ours from L3	2,830 / 13,405	1,376 / 7,420
BQSKit L3	**2,446** / 18,510	1,302 / 5,250
ours on BQSKit L3*	2,448 / 15,630	**1,298** / **5,202**

Same Qiskit L1 input, native gates and couplers. BQSKit 1.2.1 at default settings, output mapped back to the physical qubits. *On a separate BQSKit L3 run (2,459 / 18,601 and 1,301 / 5,276 before our pass).

- **Heron.** Fewer 2q gates than ours on 39/58 circuits (more on 1), about the same ESP ($\times 1.22$ vs. $\times 1.21$), but $1.38\times$ the gates of ours on L3.
- **Ion.** Fewer gates than all pipelines but our state-aware mode, and a higher ESP than ours on L3 on 45/58.
- **The price.** Approximate (median error $2 \cdot 10^{-8}$, max $1.4 \cdot 10^{-4}$), ours exact with phase. 98 s per Heron circuit on 8 workers, ours 24 s on one.
- **Combined.** Ours on BQSKit's output never adds a 2q gate or lowers ESP, removes 16 % of the Heron gates (ion 1 %) and raises ESP on 54/58, in 34 s.

Takeaway. Resynthesis finds cheaper entangling structure. Exact rewriting keeps the unitary, removes single-qubit gates, and stacks on top.

Result 4. Large circuits, 14–95 qubits

Relative to Qiskit L3

	Heron (22)			**ion** (25)	
	2q	all	ESP*	2q	all
Qiskit L1	×1.22	×1.13	×0.24	×1.40	×1.13
TKET	×1.07	×1.96	×0.27	×1.39	×1.87
ours from L1	×1.05	**×0.78**	×0.43	**×0.96**	**×0.68**
ours from L3	**×0.98**	**×0.81**	**×1.12**	×1.00	×1.00

45 Nam/VOQC circuits, 88 of 90 runs (QFT64, QFTAdd32 on Heron did not finish). pf circuits excluded, and 5 outputs whose certificates were not re-checked. *ESP over the 13 Heron circuits with L3 ESP $\geq 10^{-3}$.

Result 4. Large circuits, 14–95 qubits

Relative to Qiskit L3

	Heron (22)			**ion** (25)	
	2q	all	ESP*	2q	all
Qiskit L1	×1.22	×1.13	×0.24	×1.40	×1.13
TKET	×1.07	×1.96	×0.27	×1.39	×1.87
ours from L1	×1.05	**×0.78**	×0.43	**×0.96**	**×0.68**
ours from L3	**×0.98**	**×0.81**	**×1.12**	×1.00	×1.00

45 Nam/VOQC circuits, 88 of 90 runs (QFT64, QFTAdd32 on Heron did not finish). pf circuits excluded, and 5 outputs whose certificates were not re-checked. *ESP over the 13 Heron circuits with L3 ESP $\geq 10^{-3}$.

- **Stacked on L3, Heron.** 19 % fewer gates and 2.1 % fewer CZ. Fewer CZ on 20/22, never more, ESP ×1.12 (median ×1.14).
- **From L1.** Fewer gates than L3 everywhere but lower ESP, since L3's layout matters more than what we remove.
- **BQSKit** on the 14 comparable circuits it finished ($\leq$36 qubits). Heron 2q ×0.91 vs. ours ×0.98, but all gates ×1.35 vs. ×0.82. Its 22 outputs over 24 qubits could not be checked.

Takeaway. Larger circuits leave more for exact rewriting to remove, and the guarantees hold unchanged.

Run time

Median seconds per circuit

	60-circuit suite		large suite	
	Heron	ion	Heron	ion
Qiskit L1	–	–	0.11	0.04
Qiskit L3	0.03	0.01	0.12	0.05
TKET	0.30	0.86	8.2	9.6
BQSKit L2 (8 workers)	14.7	4.8	132*	34*
BQSKit L3 (8 workers)	53.6	11.2	842*	138*
ours from L1	10.4	3.0	**853**	187
ours from L3	6.0	4.8	117	57.1

Ours is single-threaded Python. The large suite ran several circuits at a time on eight cores. *BQSKit only on the 18 smallest large circuits.

Run time

Median seconds per circuit

	60-circuit suite		large suite	
	Heron	ion	Heron	ion
Qiskit L1	–	–	0.11	0.04
Qiskit L3	0.03	0.01	0.12	0.05
TKET	0.30	0.86	8.2	9.6
BQSKit L2 (8 workers)	14.7	4.8	132*	34*
BQSKit L3 (8 workers)	53.6	11.2	842*	138*
ours from L1	10.4	3.0	**853**	187
ours from L3	6.0	4.8	117	57.1

Ours is single-threaded Python. The large suite ran several circuits at a time on eight cores. *BQSKit only on the 18 smallest large circuits.

- Small circuits, seconds, between BQSKit L2 and L3.
- Large circuits, minutes, and up to 6.4 h for one Heron circuit. BQSKit L3 needs about 14 min per Heron circuit even on the smaller half.
- This is the main cost at scale. Non-overlapping windows could be swept in parallel, which is not done yet.

Result 5. Robust over seeds

60-circuit suite, 5 transpiler seeds (mean ± sd)

relative to Qiskit L3 (same seed)	Heron	ion
ours on L3, 2q gates	×0.975 ± 0.004	×0.998 ± 0.002
ours on L3, all gates	×0.868 ± 0.005	×0.997 ± 0.002
ours on L3, ESP	×1.03 (1.03–1.04)	×1.00
ours on L1, all gates	×0.844 ± 0.006	×0.756 ± 0.002
ours on L1, ESP	**×1.10 ± 0.11**	×1.00

1,624 outputs, all verified. The optimiser's own seed changes totals by at most 0.5 %.

Result 5. Robust over seeds

60-circuit suite, 5 transpiler seeds (mean ± sd)

relative to Qiskit L3 (same seed)	Heron	ion
ours on L3, 2q gates	×0.975 ± 0.004	×0.998 ± 0.002
ours on L3, all gates	×0.868 ± 0.005	×0.997 ± 0.002
ours on L3, ESP	×1.03 (1.03–1.04)	×1.00
ours on L1, all gates	×0.844 ± 0.006	×0.756 ± 0.002
ours on L1, ESP	**×1.10 ± 0.11**	×1.00

1,624 outputs, all verified. The optimiser's own seed changes totals by at most 0.5 %.

- Gate counts are stable to ±0.5 % whatever the seed.
- ESP from L1 ranges 0.97–1.21. It follows Qiskit's L1 layout, not our rewriting.
- The earlier ×1.21 was seed 0, the best case. The paper now reports the spread.

Result 6. What each component contributes

Heron, on top of Qiskit L3 (58 circuits)

variant	gates removed
full optimiser	**13.0 %**
no meet in the middle	12.3 %
contiguous windows	11.7 %
gate-count cost	11.6 %
no commuting shuffles	10.3 %
no resynthesis	8.2 %
window $w = 8$	7.9 %
database radius ≤ 3	6.2 %
no database	5.8 %
window $w = 4$	**1.3 %**

Result 6. What each component contributes

Heron, on top of Qiskit L3 (58 circuits)

variant	gates removed
full optimiser	**13.0 %**
no meet in the middle	12.3 %
contiguous windows	11.7 %
gate-count cost	11.6 %
no commuting shuffles	10.3 %
no resynthesis	8.2 %
window $w = 8$	7.9 %
database radius ≤ 3	6.2 %
no database	5.8 %
window $w = 4$	**1.3 %**

- **Window length** matters most, then the **database** and its radius, exactly the two knobs of Part III.
- **Resynthesis** is essential on ions. Without it the MS count from L1 rises from $\times 0.95$ to $\times 1.29$ of L3.
- Meet in the middle adds almost nothing here.
- Gate-count cost removes as many gates but gives less ESP ($\times 1.010$ vs. $\times 1.027$).

Result 7. Logical level, against VOQC and QUESO

Nam gate set $\{H, X, R_z, \mathrm{CNOT}\}$, 26 circuits

tool	gates	change	CNOT
input	10,481		4,612
TKET	11,762	$+12.2\,\%$	4,489
Qiskit L3	10,391	$-0.9\,\%$	4,511
QUESO (5 min)	9,765	$-6.8\,\%$	4,465
ours	9,480	$-9.6\,\%$	4,426
VOQC	8,417	$-19.7\,\%$	4,314
VOQC, then ours	**7,833**	$-25.3\,\%$	**4,007**

Same input files as the VOQC/Quartz/QUESO papers (from QUESO). All 159 outputs equal up to phase. QUESO had 5 min per circuit, not 1 h.

Result 7. Logical level, against VOQC and QUESO

Nam gate set $\{H, X, R_z, \mathrm{CNOT}\}$, 26 circuits

tool	gates	change	CNOT
input	10,481		4,612
TKET	11,762	$+12.2\,\%$	4,489
Qiskit L3	10,391	$-0.9\,\%$	4,511
QUESO (5 min)	9,765	$-6.8\,\%$	4,465
ours	9,480	$-9.6\,\%$	4,426
VOQC	8,417	$-19.7\,\%$	4,314
VOQC, then ours	**7,833**	$-25.3\,\%$	**4,007**

Same input files as the VOQC/Quartz/QUESO papers (from QUESO). All 159 outputs equal up to phase. QUESO had 5 min per circuit, not 1 h.

- **VOQC wins alone.** Its rotation merging over phase polynomials joins R_z gates far apart. Our windows see 3 qubits.
- **We find what it misses.** mod5_4: 35 gates vs. 56.
- **Stacked**, ours improves VOQC on 21/26 and is best overall.

Takeaway. Same lesson as with BQSKit. A global pass plus exact local cleanup beats either alone.

Where do the gains come from? (honest accounting)

Stacked on L3, summed over 58 circuits

	Qiskit L3	+ ours
Heron $\sqrt{X}/X$ pulses	7,218	6,139
Heron R_z (virtual)	5,297	4,436
Heron CZ	2,889	2,830
ion R_x/R_y pulses	3,824	2,422
ion R_z (virtual)	2,223	**3,622**
ion MS	1,378	1,376

Where do the gains come from? (honest accounting)

Stacked on L3, summed over 58 circuits

	Qiskit L3	+ ours
Heron $\sqrt{X}/X$ pulses	7,218	6,139
Heron R_z (virtual)	5,297	4,436
Heron CZ	2,889	2,830
ion R_x/R_y pulses	3,824	2,422
ion R_z (virtual)	2,223	**3,622**
ion MS	1,378	1,376

- On Heron the savings are split between pulses and R_z, plus a few CZ.
- On ions the calibrated objective *trades* pulses for free R_z. The total rises on 28/58 circuits, by design. "Never worse" holds for the objective, 2q count and ESP, **not** for the total.
- With seed 0, ESP $\times 1.21$ from L1 decomposes as $\times 1.11$ (Qiskit's L1 layout already beats its L3 layout) $\times$ $\times 1.09$ (our rewriting). We never change the layout.

Which objective? Let noisy simulation decide

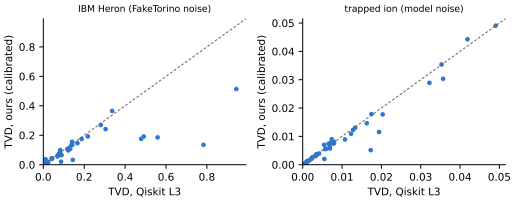
pipeline / objective	mean TVD (49 circuits) Heron	ion
Qiskit L1	0.115	0.0157
Qiskit L3	0.137	0.0091
ours, total count	0.096	0.0092
ours, 2q-weighted	**0.088**	**0.0084**
ours, calibrated $-\log F$	**0.088**	**0.0084**

Exact density-matrix probabilities (no shot noise). Heron uses FakeTorino gate errors of the actual qubits, ion a depolarising model.

Which objective? Let noisy simulation decide

pipeline / objective	mean TVD (49 circuits) Heron	ion
Qiskit L1	0.115	0.0157
Qiskit L3	0.137	0.0091
ours, total count	0.096	0.0092
ours, 2q-weighted	**0.088**	**0.0084**
ours, calibrated $-\log F$	**0.088**	**0.0084**

Exact density-matrix probabilities (no shot noise). Heron uses FakeTorino gate errors of the actual qubits, ion a depolarising model.



- Calibrated beats L3 on **39/49** (Heron) and 29/49 (ion) circuits.
- Calibrated $\approx$ 2q-weighted $>$ count $\Rightarrow$ calibrated is the default, and 2q-weighted is a good proxy without calibration data.

The other semantics, state-aware and approximate

State/measurement-aware

- Largest gains, with 2q gates $\times 0.77$ (Heron) / $\times 0.62$ (ion) relative to L3.
- Output distributions agree to $6 \cdot 10^{-12}$.
- Valid only for a complete program, with $|0 \cdots 0\rangle$ in and full measurement out.

The other semantics, state-aware and approximate

State/measurement-aware

- Largest gains, with 2q gates $\times 0.77$ (Heron) / $\times 0.62$ (ion) relative to L3.
- Output distributions agree to $6 \cdot 10^{-12}$.
- Valid only for a complete program, with $|0 \cdots 0\rangle$ in and full measurement out.

Caveat. 5 (Heron) / 7 (ion) circuits are reversible logic on $|0 \cdots 0\rangle$ with a fixed basis-state output, and they reduce to the empty circuit. Read this mode per circuit, not as an average.

Approximate ($\varepsilon = 10^{-3}$ per replacement)

- Almost no gain over exact on these suites, which have few near-identity blocks.
- Measured error always inside the tracked bound (max $6.4 \cdot 10^{-4}$).
- The exception is Nam pf2_10 (rotations $\sim 10^{-7}$ rad), 1,088 $\rightarrow$ 529 gates with error $1.3 \cdot 10^{-4} \leq$ bound $7.1 \cdot 10^{-4}$.

Every number on these slides is verified

- **Exact mode.** $\|(U - V)\psi\|$ on random states (statevector on the active qubits) is $\leq 8.2 \cdot 10^{-11}$.
- **State-aware.** The TVD of output distributions is $\leq 6 \cdot 10^{-12}$.
- **Approximate.** Measured error $\leq$ tracked bound in all runs.
- **Legality.** Native gates only, and 2q gates only on working couplers.

Every number on these slides is verified

- **Exact mode.** $\|(U - V)\psi\|$ on random states (statevector on the active qubits) is $\leq 8.2 \cdot 10^{-11}$.
- **State-aware.** The TVD of output distributions is $\leq 6 \cdot 10^{-12}$.
- **Approximate.** Measured error $\leq$ tracked bound in all runs.
- **Legality.** Native gates only, and 2q gates only on working couplers.

- **Coverage.** 250/250 paper-benchmark outputs and 720/720 hardware-suite outputs. BQSKit outputs are checked up to phase and qubit permutation (238/240, the 2 others exceed 16 qubits).
- Nam pf2_10 on Heron is routed over 20 physical qubits, beyond the main run's 16-qubit cap, so it was re-run and verified separately (same totals).
- FakeTorino has 11 dead couplers (error 1.0), removed for *all* pipelines after a first run put Qiskit L3 on them (ESP $\sim 10^{-22}$).

Verification beyond 25 qubits, replay certificates

The problem

- Statevector checks end near 25 qubits.
- A decision-diagram checker (MQT QCEC) returned no result on routed 23-qubit adders, and its simulation fallback ran out of memory.

The idea

- The optimiser logs every accepted step, the swaps of each shuffle and each window replacement.
- An independent checker replays the log. Swaps must commute, replacements must equal the old window times a scalar, all checked as small matrices on a few wires.
- The final phase repair is checked patch by patch, and the scalars must multiply to 1.

Verification beyond 25 qubits, replay certificates

The problem

- Statevector checks end near 25 qubits.
- A decision-diagram checker (MQT QCEC) returned no result on routed 23-qubit adders, and its simulation fallback ran out of memory.

The idea

- The optimiser logs every accepted step, the swaps of each shuffle and each window replacement.
- An independent checker replays the log. Swaps must commute, replacements must equal the old window times a scalar, all checked as small matrices on a few wires.
- The final phase repair is checked patch by patch, and the scalars must multiply to 1.

On the large suite

- 151/156 outputs certified, up to 309,273 steps.
- At most 3.8 s per certificate (median 0.12 s), against hours of optimisation.
- Summed residuals $< 10^{-7}$.
- 7 were rejected by the first checker, which checked the whole phase repair as one block on too many wires. 2 re-checked and accepted, 5 still to re-check.
- Uses none of the optimiser's databases or rules, and logging does not change the output.

Takeaway. Equivalence checking of the whole circuit becomes many tiny local checks, so cost grows

Part VII

Wrap-up

Lessons, limitations, critique of the original paper, and next steps.

Lesson, “exact” is easy to get subtly wrong

Found on Nam `pf1_10`/`pf2_10` (rotations of $\sim 10^{-7}$ rad), whose outputs at first verified only to 10^{-6}.

Pitfalls

- `np.allclose` has a default *relative* tolerance 10^{-5}, so near-commuting gates were swapped.
- $|\mathrm{tr}(U^\dagger V)|/d \approx 1$ is *quadratic* in the error and accepts $\|U - V\| \sim 10^{-4}$.
- Native gates reach only a *discrete* set of global phases.

Lesson, "exact" is easy to get subtly wrong

Found on Nam `pf1_10`/`pf2_10` (rotations of $\sim 10^{-7}$ rad), whose outputs at first verified only to 10^{-6}.

Pitfalls

- `np.allclose` has a default *relative* tolerance 10^{-5}, so near-commuting gates were swapped.
- $|\mathrm{tr}(U^\dagger V)|/d \approx 1$ is *quadratic* in the error and accepts $\|U - V\| \sim 10^{-4}$.
- Native gates reach only a *discrete* set of global phases.

Fixes

- Linear operator-distance tests, one tolerance 10^{-10}, `rtol=0`.
- Replacement phases snapped to the reachable set, with residuals rejected or charged to ε.
- On ions $\pm i$ is absorbed into an existing MS gate.

Takeaway. `pf1_10` now verifies to $5 \cdot 10^{-13}$ (was $1.8 \cdot 10^{-7}$). The original paper uses three different tolerances up to 10^{-3}.

Limitations of my work

- **Noise models, not hardware.** ESP and TVD come from calibration snapshots. Ion parameters are assumptions. The hardware driver is ready and tested on a fake backend, waiting for IBM access.
- **Global phase is not free for us.** Strict mode costs 240 single-qubit gates over the Heron suite. With phase as metadata (as Qiskit, TKET and BQSKit do), all gates $\times 0.838 \rightarrow \times 0.825$.

Limitations of my work

- **Noise models, not hardware.** ESP and TVD come from calibration snapshots. Ion parameters are assumptions. The hardware driver is ready and tested on a fake backend, waiting for IBM access.
- **Global phase is not free for us.** Strict mode costs 240 single-qubit gates over the Heron suite. With phase as metadata (as Qiskit, TKET and BQSKit do), all gates $\times 0.838 \rightarrow \times 0.825$.
- **Baselines.** BQSKit (approximate) needs fewer 2q gates, and VOQC beats us alone on the Nam set. Quartz was not run, QUESO had 5 min, BQSKit covers only the smaller large circuits.
- **Local only.** Layout and routing are Qiskit's, windows have ≤ 3 qubits, and the code is single-threaded Python (median 10 s per circuit on the 60-circuit suite, up to 6.4 h on large circuits, vs. $<$1 s for Qiskit).
- **Scope.** Circuits up to 95 qubits and $\sim$5,400 gates. Two 64-qubit Heron runs did not finish and 5 certificates were not re-checked. No mid-circuit measurement or feed-forward.

Weaknesses of the original paper

1. **Missing structure.** Clifford gate sets, but no tableaux and no exact keys, floating-point lookup with three tolerances, and a classifier to speed up an avoidable $O(N)$ scan.
2. **No notion of optimality.** Only compared with Qiskit, and 4.5$\times$ from an optimum computable in minutes.
3. **Equivalence.** Up to phase only (unsound under control), with no state/measurement context and no error tracking.
4. **Hardware model.** Discrete angles, all-to-all, total count. Matches neither IBM nor ion hardware.

1. **Missing structure.** Clifford gate sets, but no tableaux and no exact keys, floating-point lookup with three tolerances, and a classifier to speed up an avoidable $O(N)$ scan.
2. **No notion of optimality.** Only compared with Qiskit, and 4.5× from an optimum computable in minutes.
3. **Equivalence.** Up to phase only (unsound under control), with no state/measurement context and no error tracking.
4. **Hardware model.** Discrete angles, all-to-all, total count. Matches neither IBM nor ion hardware.
5. **Evaluation.** Random benchmarks mix, so "from 300 gates" measures input length. One circuit per table, and MATLAB vs. compiled timings.
6. **Search.** Contiguous windows, random swaps and no convergence criterion. Radius 5 is the binding limit.
7. **V3.** Unreported features and threshold, 30 % recall at default.
8. **Hardware evidence.** One 2-qubit circuit and no error bars. The effect vanishes at Qiskit level 3.
9. **Reproducibility.** Only V2 released, with visible bugs, and the database for Tables III–V unstated.

About the paper

- Reproduces, with qualifications on database, forest threshold, baselines and transpiler level.
- Its gates are Clifford, and exact optima show it is 3–4.5$\times$ from optimal.
- Theory explains why. Radius 5 and window 8 sit below the exactness threshold, and random benchmarks measure a constant.

About the paper

- Reproduces, with qualifications on database, forest threshold, baselines and transpiler level.
- Its gates are Clifford, and exact optima show it is 3–4.5× from optimal.
- Theory explains why. Radius 5 and window 8 sit below the exactness threshold, and random benchmarks measure a constant.

About the extension

- A general optimiser built on "equivalence = group action", with soundness and error bounds.
- Proven optimum on 35/40 of the paper's 3-qubit circuits without Clifford knowledge.
- On 58 real algorithms, stacked on Qiskit L3, it never adds 2q gates, never lowers ESP and removes 13 % of gates on Heron.
- BQSKit's approximate resynthesis needs fewer 2q gates. Exact rewriting keeps the unitary, removes more single-qubit gates and is faster. Run after BQSKit, it removes a further 16 % of gates on Heron.
- Up to 95 qubits, it removes 19 % of the gates on top of Qiskit L3 on Heron, verified by replay certificates.
- Stable over 5 seeds. On the Nam set VOQC then ours is the best of six tools ($-25\,\%$).

Next steps, and questions for you

Next steps

1. Layout-aware rewriting with calibrated costs (the biggest remaining lever on Heron).
2. Cheaper exact phase repair on Heron.
3. 4-qubit windows via symmetry reduction, or a learned admissible A* heuristic trained on exact tables.
4. Ground truth beyond Clifford, with exact Clifford+T normal forms for small T-counts.
5. Reach BQSKit's 2q counts and VOQC's rotation merging without approximation (larger windows, phase-polynomial passes), a Quartz baseline, and the real-hardware run (driver ready).

Next steps, and questions for you

Next steps

1. Layout-aware rewriting with calibrated costs (the biggest remaining lever on Heron).
2. Cheaper exact phase repair on Heron.
3. 4-qubit windows via symmetry reduction, or a learned admissible A* heuristic trained on exact tables.
4. Ground truth beyond Clifford, with exact Clifford+T normal forms for small T-counts.
5. Reach BQSKit's 2q counts and VOQC's rotation merging without approximation (larger windows, phase-polynomial passes), a Quartz baseline, and the real-hardware run (driver ready).

Where I'd like your input

1. Is the target a short note on the reproduction + optimality gap, or a full paper on the general framework?
2. Is real-hardware access available for an error-barred experiment?
3. Which venue fits, a quantum-computing one (theory + benchmarks) or a compiler venue?
4. Should we contact the authors about the unstated database and the code bugs?

Deliverables

What	Where
Study notes (read instead of the paper)	`manuscripts/notes/notes.pdf`
Reproduction report (every table and figure)	`manuscripts/reproduction/reproduction_report.pdf`
Extension paper	`manuscripts/extension/extension_paper.pdf`
Python port, exact Clifford engine, general optimiser	`qcr/`
Experiment scripts and raw results	`experiments/`, `results/`
This talk	`manuscripts/talk/talk.pdf`

Thank you

Backup

Backup. Paper-benchmark numbers (mean total gate count)

set	Qiskit L3	paper's method	Clifford CW-sem	optimum / bound	general
T3 (ion, 3q, 20)	127.2	47.1	–	10.4	**10.8**
T4 (IBM-style, 3q, 20)	104.0	36.8	–	12.2	**12.5**
journal ion (4q, 100)	164.9*	78.8*	45.6	$\geq$11.1	**46.1**
journal IBM-style (4q, 100)	140.9*	76.0*	49.3	$\geq$12.9	**51.4**
T5 (IBM-style, 15q, 10)	332.1	335.9	324.6	–	**327.7**

*first 30 circuits. The general engine uses up-to-phase semantics (as in the paper), total count, discrete $\pm\pi/2$ alphabets and all-to-all connectivity.

Backup. Minimal entangling-gate counts, 3 qubits

entangling gates	0	1	2	3	4	5	6
3-qubit Cliffords	13,824	373,248	5,971,968	38,486,016	42,052,608	5,971,968	27,648

Mean 3.51, maximum 6. Identical for both alphabets ($R_{XX}(\frac{\pi}{2})$ and CZ are locally equivalent). For 2 qubits the counts are 576 / 5,184 / 5,184 / 576 for 0–3, the known CNOT-count distribution.

Backup. Worked examples from the reproduction

- **Fig. 1.** The 8-gate and 4-gate Bell circuits both equal CNOT $(H \otimes I)$ to 10^{-13}, and no 3-gate circuit exists.
- **Fig. 4 analogue** (2 qubits, 40 gates). $40 \to 34 \to 26 \to 24 \to 20 \to 18 \to 16 \to 8 \to 4$, and every step removes an even number of gates (parity theorem). The paper's final circuit contains an inverse pair, and the optimum is 2.
- **Fig. 5 analogue** (3 qubits, 200 gates). $200 \to 36$ in 31 replacements.
- **Authors' demo circuit** (5 qubits, 300 gates). Authors 140, our port 140 (132–148) with the demo database and 115 (112–117) with the 24-op depth-5 database.